

Problem statement

Ingredients: we can use

- A ML model $\hat{m}$, a **loss function** ℓ , and a **feature of interest** X ,
- True conditional distribution** $P_{X|Z}^*$ (MX assumption) – or good estimate,
- The ability to draw samples $\tilde{X} \sim P_{X|Z}^*(z)$ for any z .

Main goal: obtain a sequential test for the null hypothesis

$$\mathcal{H}_{P_{X|Z}^*}^{\text{MX}} := \{P(X, Y, Z) : Y \perp_P X | Z, P_{X|Z} = P_{X|Z}^*\},$$

- Which is **anytime-valid** whenever $P_{X|Z}^*$ is known exactly,
- Which is **powerful** under a wide range of alternatives Q ,
- Which can make use of **any machine learning model** $\hat{m}$.

E-variables background

- $S \geq 0$ is an **e-variable** for $\mathcal{H}_0$ if $\mathbb{E}_P[S] \leq 1, \forall P \in \mathcal{H}_0$.
- $(E_n)_n$ is an **e-process** if E_τ is an e-variable for any stopping time τ .
- We can reject $\mathcal{H}_0$ as soon as $E_n > 1/\alpha$.
- Type-I error:** by Ville's inequality:

$$P\left(\exists n : E_n \geq \frac{1}{\alpha}\right) \leq \alpha, \forall P \in \mathcal{H}_0.$$

- Growth-rate optimal (GRO):** e-variable S maximizing $\mathbb{E}_Q[\log(S)]$ (**e-power**).

$$\prod_{i=1}^n S_i = \exp\left(\sum_{i=1}^n \log(S_i)\right) \approx \exp(n\mathbb{E}[\log(S)]).$$

Candidate I: Exponential e-variable

The **GRO e-variable** [Grünwald et al., 2024] for testing $\mathcal{H}_{P_{X|Z}^*}^{\text{MX}}$ vs $\{Q\}$ is:

$$\frac{f_{Y|X,Z}(Y|X, Z)}{f_{Y|Z}(Y|Z)} = \frac{f_{Y|X,Z}(Y|X, Z)}{\mathbb{E}_{\tilde{X}|Z}[f_{Y|X,Z}(Y|\tilde{X}, Z)]},$$

where Q has density $f(x, y, z)$ and $\tilde{X} \sim P_{X|Z}^*(Z)$.

Practical limitation: Hard to estimate $f_{Y|X,Z}$.

We propose:

$$E_{\text{exp}}^\eta := \frac{e^{-\eta \ell(\hat{m}(X, Z), Y)}}{\mathbb{E}_{\tilde{X}|Z}[e^{-\eta \ell(\hat{m}(\tilde{X}, Z), Y)}]},$$

with a free parameter $\eta \geq 0$.

- Good **approximation of the GRO** if: $\ell(m(x, z), y) \approx -\eta \log(f_{Y|X,Z}(y|x, z))$.
- E.g. in regression with ℓ^2 -loss and Gaussian noise, and classification with CE.

Testing Algorithm

While $\prod_{i=1}^n E_i \leq \frac{1}{\alpha}$ **do**

- Sample $(X_{n+1}, Z_{n+1}, Y_{n+1})$.
- Update the model to $\hat{m}_n$ using all past data $\{(X_j, Z_j, Y_j)\}_{j=1}^n$ or an online step.
- Perform a prequential parameter update:

$$\hat{\eta}_n = \arg \max_{\eta} \left(\prod_{i=1}^n E_{\text{exp}}^\eta(X_i, Y_i, Z_i) \right)$$

- Compute

$$E_{n+1} = E_{\text{exp}}^{\hat{\eta}_n}(X_{n+1}, Y_{n+1}, Z_{n+1}).$$

Antisymmetric e-variables

To leverage ML models, Shaer et al. [2023] introduce **importance statistics**:

$$l := \ell(\hat{m}(X, Z), Y) \quad \text{and} \quad \tilde{l} := \ell(\hat{m}(\tilde{X}, Z), Y),$$

and e-variables of the form:

$$1 + g(l, \tilde{l}),$$

with **antisymmetric** functions g ($g(a, b) = -g(b, a)$) such that $g(a, b) \geq -1$.

- Simplifies conditional independence question into **exchangeability**

$$Y \perp X | Z \longrightarrow (l, \tilde{l}) \stackrel{d}{=} (\tilde{l}, l).$$

- Examples:

$$\underbrace{1 + \lambda \text{sign}(\tilde{l} - l)}_{\text{Sign}}, \quad \underbrace{1 + \lambda \tanh(\tilde{l} - l)}_{\text{Tanh}}, \quad \underbrace{1 + \lambda(\tilde{l} - l)}_{\text{Coin-betting (bounded loss)}}.$$

Practical limitation: Function g is left unspecified, unclear how to choose it.

Exponential vs antisymmetric e-power

- If $\ell(m(x, z), y) = -\eta \log(f_{Y|X,Z}(y|x, z))$, E_{exp}^η is GRO (dominates all antisymmetric).
- Do antisymmetric e-variables ever dominate?

Thm: Exponential vs Coin-betting e-power

Define *predictiveness signal*:

$$\Delta := \mathbb{E}[\ell(m(\tilde{X}, Z), Y)] - \mathbb{E}[\ell(m(X, Z), Y)] \geq 0,$$

and loss variances:

$$\sigma_0^2 := \mathbb{V}(\ell(m(\tilde{X}, Z), Y)), \quad \sigma_1^2 := \mathbb{V}(\ell(m(X, Z), Y)).$$

Optimal e-powers for E_{exp}^η and E_{cb}^λ are:

$$\mathbb{E}[\log E_{\text{exp}}^{\eta*}] = \frac{\Delta^2}{2\sigma_0^2} + O(\Delta^3), \quad \mathbb{E}[\log E_{\text{cb}}^{\lambda*}] = \frac{1}{2} \frac{\Delta^2}{\sigma_1^2 + \Delta^2} + O(\Delta^3).$$

$\rightarrow$ If $\frac{\sigma_0^2}{\sigma_1^2 + \Delta^2} > 1$, **Coin-betting can be more e-powerful!**

Candidate II: Antisymmetric KDE e-variable

$$E_{\text{KDE}}^{\hat{f}} := 1 + \mathbb{E}_{\tilde{X}|Z} \left[\frac{\hat{f}(l, \tilde{l}) - \hat{f}(\tilde{l}, l)}{\hat{f}(l, \tilde{l}) + \hat{f}(\tilde{l}, l)} \right] = \mathbb{E}_{\tilde{X}|Z} \left[\frac{2\hat{f}(l, \tilde{l})}{\hat{f}(l, \tilde{l}) + \hat{f}(\tilde{l}, l)} \right],$$

where $\hat{f}$ approximates the density of $(l, \tilde{l})$ via Kernel Density Estimation.

- The GRO e-variable for testing exchangeability null $(A, B) \stackrel{d}{=} (B, A)$ against $Q(A, B)$ with density $f(a, b)$ is:

$$\frac{2f(A, B)}{f(A, B) + f(B, A)}.$$

- For good approximations $\tilde{f} \approx f$ and any antisymmetric g :

$$\mathbb{E} \left[\log \left(\frac{2\hat{f}(l, \tilde{l})}{\hat{f}(l, \tilde{l}) + \hat{f}(\tilde{l}, l)} \right) \right] \geq \mathbb{E} \left[\log (1 + g(l, \tilde{l})) \right].$$

- We simplify the null to $(l, \tilde{l}) \stackrel{d}{=} (\tilde{l}, l) + \text{Model-X assumption}$.
- $E_{\text{KDE}}^{\hat{f}}$ approximates the GRO **among antisymmetric** e-variables!

$\rightarrow$ **Approximate the antisymmetric GRO** with KDE and derandomize $\tilde{X}$.

Experiments

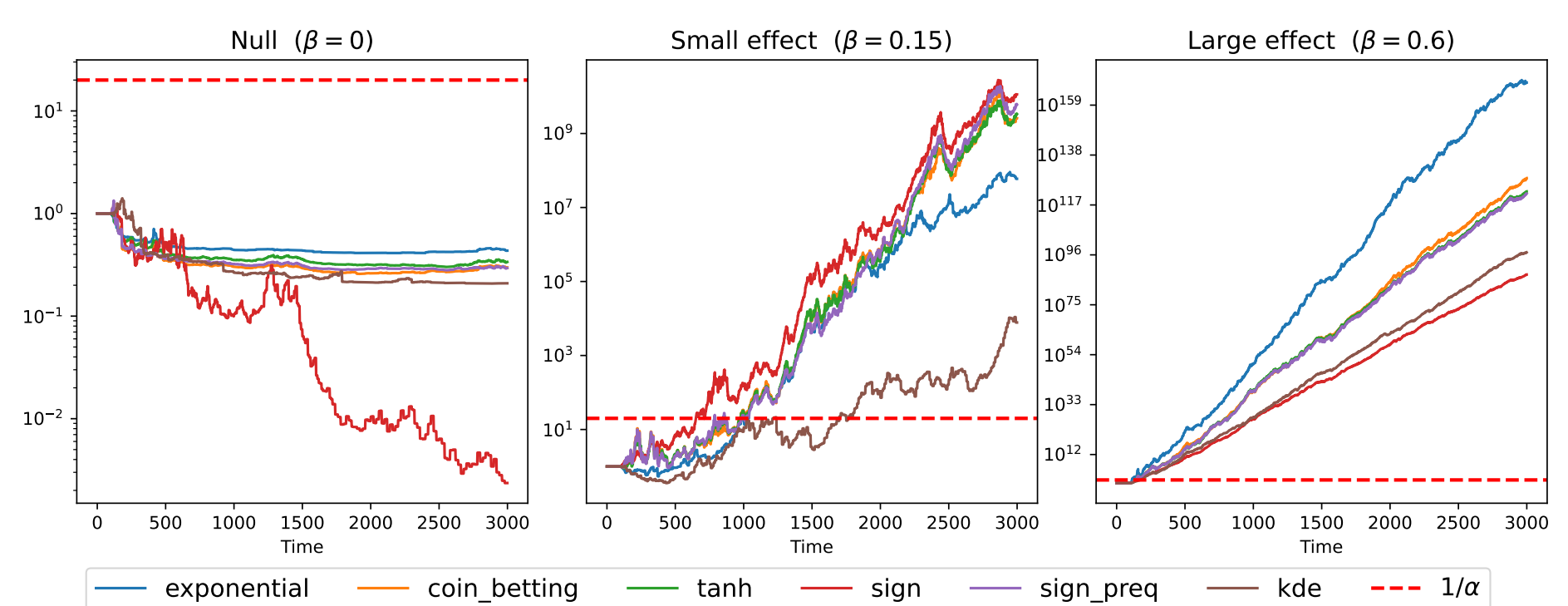


Figure: **Simulated data.** $y = (Z^\top W)^2 + \beta X + \varepsilon$, where $X | Z \sim \mathcal{N}(U^\top Z, 1)$ and $Z \sim \mathcal{N}(0, I_d)$ with $d = 19$.

- Under the null ($\beta = 0$): neither process rejects.
- For a weak signal ($\beta = 0.15$): antisymmetric e-variables beat the exponential.
- For a stronger signal ($\beta = 0.6$): the exponential exhibits fastest growth.

Conclusions

- We make the GRO e-variable practical through the exponential e-variable.
- Under misspecification, antisymmetric e-variables can dominate.
- We propose the GRO antisymmetric based on estimating a 2-d density.
- Ongoing work:** we are studying:
 - The KDE e-variable and the error induced by estimating density $\hat{f}$.
 - Safety guarantees when the Model-X distribution $P_{X|Z}^*$ is itself estimated.

References

- P. Grünwald, A. Henzi, and T. Lardy. Anytime-valid tests of conditional independence under model-x. *JASA*, 2024.
- S. Shaer, G. Maman, and Y. Romano. Model-x sequential testing for conditional independence via testing by betting. In *AISTATS*, 2023.